

When Does Multimodal Diffusion Inverse Kinematics Help Soft Continuum Arms—and Does It Survive Model Mismatch?

Adhvay Jagadeesh

Abstract—Soft continuum arms are highly redundant: many actuation patterns realize the same tip position. Deterministic learned inverse kinematics (IK) regressors average these incompatible solutions into invalid ones, while a conditional diffusion model can represent the full solution distribution. We present a controlled simulation study that asks two questions the existing literature leaves open: *when* does this multimodality actually help, and does the advantage survive a change of dynamics model? On a piecewise-constant-curvature (PCC) pneumatic arm, a conditional diffusion model matches a deterministic baseline where redundancy is absent and dominates where it is present (mode recall 0.84 ± 0.03 vs. exactly 0; obstacle-avoidance success 97.8% vs. 4.0%; three seeds), and matches a strong optimization-based solver on accuracy at a tenth of its query cost without needing the simulator at query time. Two controlled manipulations show the advantage is specifically a redundancy effect: conditioning on the whole arm shape instead of the tip collapses sampled diversity by $\sim 85\times$ and shrinks the baseline’s disadvantage from $\sim 30\times$ to $\sim 3\times$, while increasing the number of segments widens the gap. However, when actuations from the PCC-trained model are executed on a high-fidelity Cosserat-rod simulator, accuracy collapses on every seed ($0.67\text{ mm} \rightarrow 139.9\text{ mm}$, 0% success): the learned inverse map is simulator-specific. We characterize what recovers this loss: candidate selection under a query budget helps but is bounded; a diversity-aware query order significantly outperforms a confidence order (paired $t=3.3$); guidance-based adaptation yields a modest, statistically suggestive gain; gain-domain-randomization fails entirely; and fine-tuning on target-domain data reveals a sharp retention–transfer cliff with no data-mixing sweet spot. All code, data, and per-experiment result files are made available.

Index Terms—Modeling, Control, and Learning for Soft Robots; Deep Learning Methods; inverse kinematics; sim-to-sim transfer.

I. INTRODUCTION

A MULTI-SEGMENT pneumatic continuum arm with three chambers per segment has more actuation degrees of freedom than task degrees of freedom: for a four-segment arm, a 12-D actuation vector determines a 3-D tip position. The inverse map is therefore one-to-many, and its preimages are not merely a connected null space—distinct arm shapes reach the same tip. A deterministic learned IK regressor trained by motor babbling [1] fits the conditional *mean* of this multimodal distribution and can return an actuation that realizes none of the true solutions. A conditional diffusion model [2], [3] instead learns the full conditional distribution $p(\mathbf{q}|\text{target})$ and can sample all solution modes.

A. Jagadeesh is an independent researcher (e-mail: mail.ack.ventures@gmail.com).

Diffusion models for robot action generation are now well-established [4], and diffusion-based IK has been demonstrated for rigid multi-arm systems [5]. What the literature does not yet provide for *soft continuum* arms is (i) a controlled account of when the multimodality actually matters—as opposed to diffusion simply being a strong regressor—and (ii) an honest measurement of whether the learned inverse map survives the model mismatch that any simulation-trained soft-arm controller must eventually face. Soft arms make question (ii) unusually pressing: their dynamics are hard to model, so the gap between any two models of the *same* physical arm is large, and a method whose advantage evaporates under that gap is of limited practical value.

We answer both questions in a fully reproducible sim-to-sim setting: an idealized, fast PCC model [6] as the training simulator, and a Cosserat-rod simulation (PyElastica [7], [8]) of the same arm as the high-fidelity evaluation domain. Our contributions:

- 1) **A validated advantage.** With three-seed statistics, diffusion IK dominates learned MLP and mixture-density-network (MDN) baselines, recovers 0.84 ± 0.03 of ground-truth IK modes where they structurally recover none, and matches a classical random-restart optimization solver on accuracy at a tenth of its query cost without simulator access at query time—translating into a 97.8% vs. 4.0% obstacle-avoidance gap over the learned baselines (Sec. IV).
- 2) **The mechanism, isolated.** Two controlled manipulations of redundancy—conditioning on the whole body shape, and varying the number of segments—show the advantage is specifically redundancy resolution, not generic model capacity (Sec. V).
- 3) **A quantified transfer collapse—and what recovers it.** The learned inverse map collapses under simulator change on every seed. We quantify the recovery achievable by inference-time selection and guidance, show a diversity-aware query ordering significantly outperforms the obvious confidence ordering (paired $t=3.3$, $p \approx 0.002$), show gain-domain-randomization achieves nothing, and map a sharp retention–transfer cliff for target-domain fine-tuning (Sec. VI).

Negative and bounded results are reported with the same prominence as positive ones; Sec. VII states the limitations explicitly.

II. RELATED WORK

IK for continuum arms. Non-learned approaches solve continuum IK by optimization through a model at query time: Jacobian-based schemes for nonconstant-curvature arms [1] and real-time FEM inversion [9]; the constant-curvature kinematics underlying our fast simulator are reviewed in [6], and STIFF-FLOP-style pneumatic arms motivate the three-chamber actuation model [10]. We include an optimization-based baseline of this family (random-restart gradient IK through the differentiable model). Learned approaches fit a deterministic target-to-actuation map on motor-babbling data [1]; mixture density networks [11] are the classical answer to multimodal regression and serve as the learned multimodal baseline.

Diffusion models for robots. Denoising diffusion models [2] with DDIM sampling [3] and classifier-free guidance [12] underpin diffusion policies [4]. IKDiffuser [5] solves IK for *rigid multi-arm* systems with flexible kinematic-tree topology; physically compliant continuum bodies are outside its scope. Selecting one action from a diffusion policy’s samples via learned scorers is an active area; our transfer-guided sampling instead injects the score’s gradient *during* generation, in the spirit of classifier guidance [13], and our query-budget framing asks how many *expensive target-domain evaluations* selection needs, which we have not seen posed for soft-arm IK.

Crossing the reality gap. Domain randomization [14] and target-domain fine-tuning are the standard levers; catastrophic forgetting under naive fine-tuning is well-documented [15]. We evaluate both levers in a controlled sim-to-sim instance and find the first fails completely while the second exposes a sharp retention–transfer tension.

III. SETUP

Arm and training simulator. The training domain is a four-segment PCC arm (segment length 60 mm, total length 240 mm). Each segment has three 120°-spaced chambers with normalized pressures $\mathbf{q} \in [0, 1]^{12}$; a linear chamber-to-curvature map (gain 25) yields per-segment curvature components (k_x, k_y) , and forward kinematics composes constant-curvature arcs. The model is batched and differentiable.

Data and models. Training data are 5×10^5 i.i.d. uniform actuations pushed through forward kinematics. The diffusion model is a residual MLP denoiser (six FiLM-conditioned blocks, width 512) trained as a DDPM ($T=1000$, cosine schedule) with classifier-free guidance (condition dropout 0.1); sampling uses DDIM. Ablations (Sec. IV) fix guidance weight 1.5 and 20 DDIM steps. Baselines: an MLP regressor and an MDN (eight mixture components), matched in width, plus a classical optimization-based reference—gradient-descent IK through the differentiable PCC model from 32 random restarts (Adam, 300 iterations), denoted grad-IK. Unless stated otherwise we sample $K=32$ candidates per target and report best-of- K ; success means tip error under 5 mm (2% of arm length). The obstacle task (E3) runs 200 trials, each placing three random spherical obstacles (radius 2 cm) in the workspace;

TABLE I

CORE COMPARISON ON THE PCC ARM (MEAN $\pm$ STD, 3 SEEDS; $K=32$). MODE RECALL AND DIVERSITY ARE COMPUTED OVER SUCCESSFUL SAMPLES ONLY. GRAD-IK IS A SINGLE OPTIMIZATION RUN PER TARGET (NO TRAINING, HENCE NO SEEDS) AND REQUIRES THE DIFFERENTIABLE SIMULATOR AT QUERY TIME.

	Diffusion	grad-IK	MLP	MDN
Tip err. best-of- K (mm)	0.44 ± 0.03	0.23	13.4 ± 0.0	17.3 ± 0.2
Success @ 5 mm	1.00	0.994	0.060	0.027
Mode recall	0.84 ± 0.03	0.73	0.00	0.00
Diversity (curv. space)	28.2 ± 0.2	9.5	0.0	0.0
Obstacle success	0.978 ± 0.002	0.960	0.040	0.018
Query time (ms/target)	38.6 ± 4.9	395	0.013	0.022

success requires any of the K samples to be simultaneously accurate and collision-free.

Ground-truth modes. For evaluation targets we enumerate the true IK solution set by a 2×10^6 -sample sweep, keeping actuations whose tip lands within tolerance and clustering with DBSCAN [16] ($\epsilon=12$, $\text{min_samples}=5$; a generated sample matches a mode within radius 15). Clustering, mode matching, and diversity are computed in the 8-D per-segment *curvature* space (k_x, k_y) rather than raw pressures: the chamber-to-curvature map has a per-segment null space (common-mode pressure offsets change $\mathbf{q}$ but not the shape), which would otherwise contaminate distance-based metrics. Evaluation targets average 2.05 ground-truth modes (range 0–6); targets where the sweep finds no mode are excluded from recall averages.

High-fidelity domain. The transfer domain is a Cosserat-rod simulation of the same arm in PyElastica [7], [8]: one rod, 48 elements, per-segment constant internal couples derived from the same chamber pressures, damped to quasi-static equilibrium (~ 4 s of compute per evaluation). The two simulators agree exactly in the zero-curvature case—uniform pressures produce the identical straight configuration (tip at $[0, 0, 0.24]$ m) in both—so the discrepancy lies in the bending response, not in trivial misalignment. A sweep over the couple gain further confirmed it is directional/structural rather than a scale mismatch: no gain setting aligns the two workspaces.

IV. WHEN MULTIMODALITY HELPS

Table I summarizes the four core experiments. **E1 (accuracy):** best-of- K diffusion reaches 0.44 mm mean tip error with 100% success; the deterministic MLP, structurally limited to the conditional mean, plateaus at 13.4 mm, and the MDN does not improve on it. The classical grad-IK reference reaches 0.23 mm (99.4%)—when the true model is available to optimize through at query time, classical optimization is fully competitive on accuracy. **E2 (multimodality):** against enumerated ground-truth solution sets, diffusion recovers 0.84 ± 0.03 of modes; both baselines recover none—the MLP because it is deterministic, and the MDN not through classic component collapse but because no component is accurate enough: mode recall and diversity are computed over *successful* samples only, and the MDN almost never produces one. Per-component analysis rules out a training pathology—NLL decreases monotonically at every epoch, no variance reaches its clamp floor, and 7.8 of 8 components remain effective—yet even the *best* component mean per target carries 7.5 mm error (median

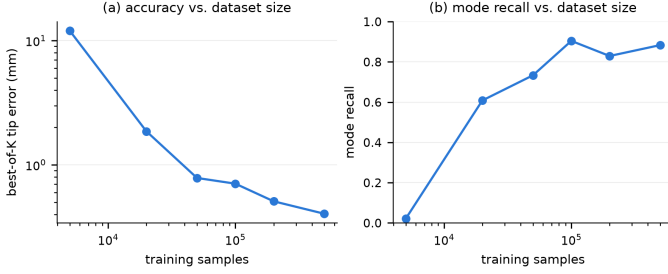


Fig. 1. Dataset-size ablation (single seed). (a) Best-of- K tip error and (b) mode recall vs. training-set size; both saturate by $\sim 0.5\text{--}1 \times 10^5$ samples.

component 9.2 mm) against the 5 mm tolerance, and sampling with the learned σ smears this further. Nor is it a tuning artifact: sweeping the component count over $\{4, 8, 16, 32\}$ improves best-of- K error only from 18.7 to 14.8 mm—never surpassing the plain MLP—and mode recall remains exactly zero at every setting. grad- IK ’s random restarts, by contrast, *do* recover modes (recall 0.73)—multimodality does not require learning—though with a third of diffusion’s solution diversity (9.5 vs. 28.2). **E3 (downstream value)**: when success requires *any* of K samples to be both accurate and collision-free among random obstacles, diffusion succeeds on 97.8% of trials vs. 4.0% (MLP)—redundancy is only exploitable if the model can represent it. **E4 (cost)**: 38.6 ms per target for a batch of 32 samples on CPU, versus 395 ms for grad- IK —and, more fundamentally, grad- IK requires the differentiable simulator in the loop at every query, whereas the learned methods answer from samples of the plant alone. That amortization property is exactly what Sec. VI puts under stress. With three seeds, formal hypothesis testing adds little; the separations are instead unambiguous by inspection: on every metric in Table I the *worst* diffusion seed outperforms the *best* baseline seed—for tip error, 0.49 vs. 13.33 mm, a gap exceeding $400\times$ the across-seed standard deviation.

Three ablations support the protocol: guidance weight has an inverted-U optimum at 1.5 (both accuracy and diversity collapse beyond ~ 3); accuracy and recall saturate by ~ 20 DDIM steps (a $3\times$ speedup over 50 steps at no measurable cost); and performance saturates by $\sim 0.5\text{--}1 \times 10^5$ training samples (Fig. 1).

V. THE ADVANTAGE IS REDUNDANCY, SPECIFICALLY

If the advantage in Table I is redundancy resolution, it should shrink when redundancy is removed and grow when redundancy is added. Both predictions hold (Fig. 2).

Removing redundancy: shape conditioning. Conditioning the same models on the full 24-D body shape (eight backbone points) instead of the 3-D tip pins down almost the entire configuration. Diffusion’s sampled diversity collapses from 28.16 ± 0.22 to 0.346 ± 0.022 ($\sim 85\times$), and the MLP’s accuracy disadvantage shrinks from $\sim 30\times$ (13.4 vs. 0.44 mm) to $\sim 3\times$ (0.49 vs. 0.16 mm), with both models at 100% success (3 seeds). When there is one answer, a deterministic regressor is nearly as good; diffusion’s edge is specifically the multimodal case.

TABLE II
RECOVERY MECHANISMS AT A GLANCE (ELASTICA-DOMAIN PERFORMANCE; BEST CONFIGURATION OF EACH MECHANISM).

Mechanism	err (mm)	succ.	Cost / caveat
none (PCC-trained)	139.9 ± 1.8	0%	—
best-of-32 selection	130.4	0%	32 target queries/target
+ diversity order	-6.7 at $m=8$	0%	same queries; $p \approx 0.002$
guided sampling	129.0	0%	400 labels; suggestive
gain randomization	140.6	0%	degrades source; no gain
fine-tune, 0% mix	67.1 ± 1.0	10%	forgets source (133 mm)
fine-tune, $\geq 5\%$ mix	≥ 133.8	$\leq 10\%$	retains source, forfeits gain

Adding redundancy: morphology. Training fresh models on two- and six-segment variants of the same arm family (tolerance scaled to 2% of arm length), the MLP’s error grows from 1.7 mm (2 seg, 75% success) to 33.9 mm (6 seg, 1.2% success) while diffusion degrades gracefully (0.11 to 0.92 mm, $\geq 99.8\%$ success); the obstacle-task gap at six segments is 100% vs. 1%—the largest in the study. More redundancy means worse mode averaging and more collision-free alternatives that only a multimodal model can offer.

VI. DOES IT SURVIVE MODEL MISMATCH?

A. The transfer collapse

Executing the PCC-trained diffusion model’s actuations on the Cosserat-rod simulator collapses accuracy from 0.67 ± 0.03 mm (100% success) to 139.9 ± 1.8 mm with 0% success on every seed ($n=20$ held-out targets, $K=8$); the MLP collapses similarly (13.8 $\rightarrow$ 164.4 mm). The learned inverse map is simulator-specific. The rest of this section quantifies what does—and does not—recover the loss; Table II summarizes.

B. Selection under a query budget

A multimodal model offers something a deterministic one cannot: K diverse candidates to test against the target domain. Spending a budget of m Elastica evaluations and keeping the best candidate reduces mean error from 161.5 ($m=1$) to 130.4 mm ($m=32$), improving 44 of 45 targets (Fig. 3)—real and broad-based, but to be blunt about its practical value: success stays at 0%, the gap to tolerance remains $\sim 26\times$, and under this budget selection never actually solves a single target. *How* the budget is spent matters: a greedy farthest-point ordering in curvature space (seeded from the same most-confident candidate, so $m=1$ is identical) beats confidence ordering by 6.7 mm at $m=8$ (paired $t=3.34$, $p \approx 0.002$; 24/12 targets) and 3.3 mm at $m=16$ ($t=3.21$). Notably, neither candidate diversity itself nor local PCC Jacobian sensitivity predicts *which* targets benefit: at $n=45$ —sized to detect $r \geq 0.41$ at 80% power, i.e., the effect size an $n=20$ pilot had suggested—the observed correlations are $r=0.20$ ($p=0.19$) and $r=0.02$ ($p=0.88$). This rules out the pilot’s effect but not smaller ones; the pilot’s non-replication is a caution we report rather than hide.

C. Guiding generation toward the target domain

Beyond selecting among candidates, we steer generation itself: a small regressor trained on 400 Elastica-labeled candidates (25 targets) predicts transfer error from (candidate,

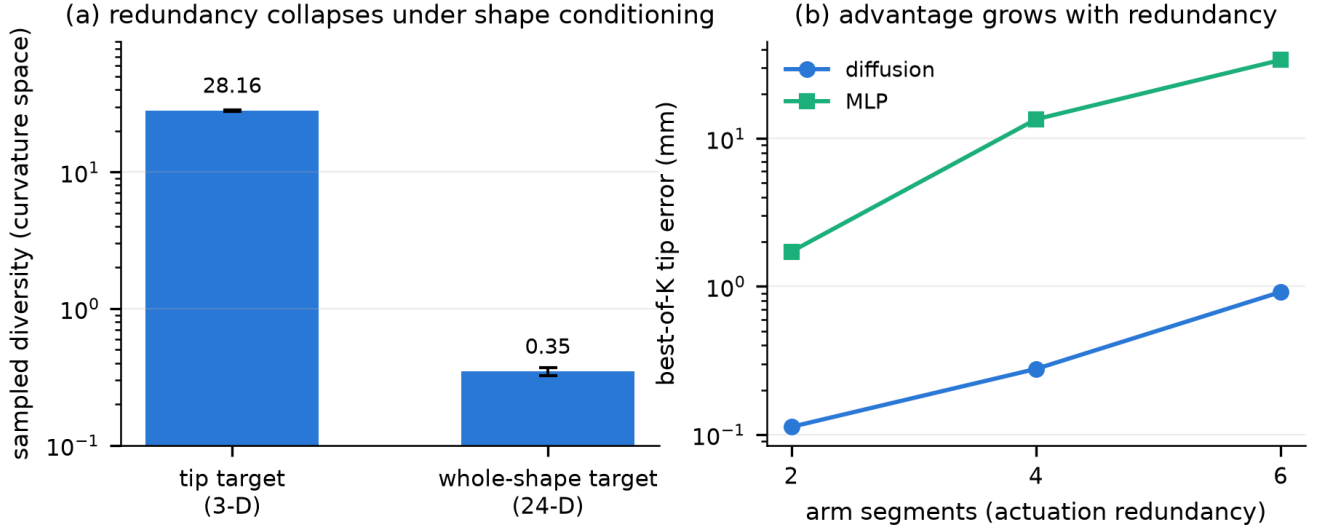


Fig. 2. The diffusion advantage tracks actuation redundancy. (a) Replacing the 3-D tip target with a 24-D whole-shape target removes most of the redundancy: sampled diversity collapses $\sim 85\times$ (28.16 ± 0.22 to 0.346 ± 0.022 ; mean $\pm$ std, 3 seeds). (b) Increasing the number of segments (single seed per morphology) widens the diffusion–MLP accuracy gap; at six segments the obstacle-task gap reaches 100% vs. 1%.

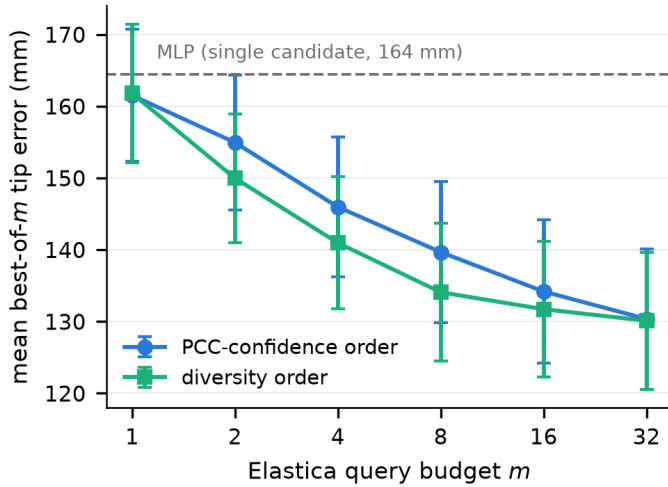


Fig. 3. Candidate selection under an Elastica query budget ($n=45$ targets, $K=32$ candidates, mean $\pm$ SEM). Curves shown are independent runs of the two orderings; the statistical comparison in the text uses a *paired* design (one candidate set per target, both orderings reindex the same cached per-candidate errors), which is significant at $m=8$ and $m=16$ ($t=3.34$ and 3.21 , $p \approx 0.002$).

target), and its gradient nudges the denoised estimate at every DDIM step, as in classifier guidance [13]. On 15 *held-out* targets this improves best-of-16 error from 134.6 to 129.0 mm (paired 5.6 ± 2.5 mm, $t=2.27$, $p \approx 0.04$; 11/15 targets). This is generalization to unseen targets, but it is the weakest effect in the paper: all p -values here are uncorrected, and across the family of $\sim$ six hypothesis tests we report, only the ordering result ($p \approx 0.002$) survives a Bonferroni correction—we therefore read this one as *suggestive* rather than confirmed. Combining it with diversity-ordering adds nothing: guidance *concentrates* the candidate distribution while diversity-ordering *spreads* selection, and a 2×2 factorial shows the mechanisms in tension rather than compounding.

TABLE III
FINE-TUNING ON 10^4 ELASTICA-SIMULATED SAMPLES AT SIX PCC-DATA MIXING RATIOS (MEAN $\pm$ STD, 3 SEEDS; HELD-OUT TARGETS).

PCC mix	PCC err (mm)	succ.	Elastica err (mm)	succ.
none (baseline)	0.67 ± 0.03	100%	139.9 ± 1.8	0%
0%	133.4 ± 1.2	0%	67.1 ± 1.0	10%
5%	30.7 ± 0.6	6%	133.8 ± 3.1	8%
10%	21.7 ± 0.5	17%	143.2 ± 4.9	8%
15%	16.7 ± 0.2	26%	146.0 ± 3.4	8%
20%	11.6 ± 0.3	51%	148.0 ± 1.6	10%
50%	3.0 ± 0.4	96%	152.3 ± 0.4	3%

D. Training-time levers

Domain randomization fails. Randomizing the curvature gain per-sample ($\mathcal{U}(15, 35)$) during data generation costs PCC accuracy ($0.44 \rightarrow 2.18$ mm) and yields *zero* transfer improvement ($140.6 \rightarrow 140.6$ mm)—consistent with the mismatch being directional rather than magnitude-based: randomizing how much the arm bends cannot fix a discrepancy in which direction it bends.

Target-domain data works—at a price.

Fine-tuning the pretrained model *exclusively* on 10^4 Elastica-simulated samples yields the only nonzero transfer success of any method— 67.1 ± 1.0 mm and 10% success (2/20 targets on each seed; recall the 5% metric granularity)—at the cost of catastrophic forgetting [15] of the source domain ($0.67 \rightarrow 133.4$ mm). Mixing PCC data back into fine-tuning batches restores retention smoothly and monotonically across all six ratios ($133.4 \rightarrow 30.7 \rightarrow 21.7 \rightarrow 16.7 \rightarrow 11.6 \rightarrow 3.0$ mm), but the transfer gain does not degrade in step: the fine sweep *locates* the collapse below the first grid point—at just 5% PCC data, transfer error has already rebounded from 67.1 to 133.8 ± 3.1 mm, forfeiting $\sim 92\%$ of the improvement over the no-fine-tune baseline, and it drifts only slowly thereafter (Table III, Fig. 4). The values at intermediate ratios are stable across seeds (std ≤ 4.9 mm), so this is not an unlucky training

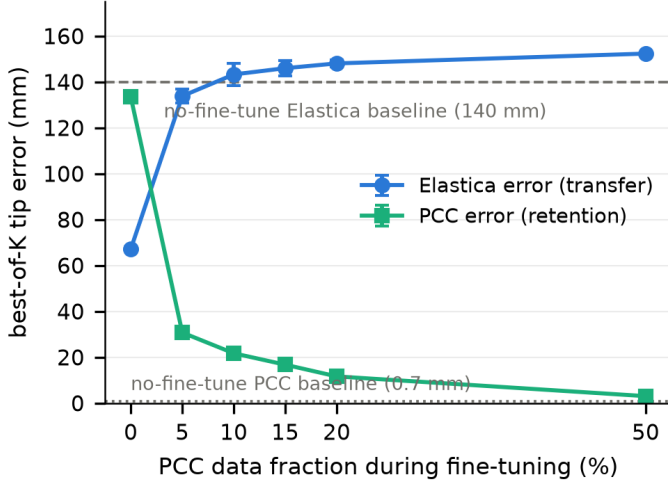


Fig. 4. The retention–transfer cliff (mean $\pm$ std, 3 seeds; six mixing ratios). Fine-tuning the PCC-pretrained model on 10^4 Elastica samples: retention (PCC error) improves smoothly and monotonically as PCC data is mixed back in, but the transfer gain collapses almost entirely by the first grid point (5% PCC data). There is no data-mixing sweet spot.

run. Retention and transfer sit on a *cliff* whose edge lies below 5% mixing, not on a smooth tradeoff curve—under per-batch data mixing there is no sweet spot, suggesting parameter-level regularization (e.g. elastic weight consolidation [15]) as the necessary next mechanism. (Binary success has 5% granularity at $n_{\text{targets}}=20$; mean error carries the statistical weight throughout.)

VII. LIMITATIONS

(1) **Simulation-only.** Both domains are simulators; this is a controlled sim-to-sim study, and no claim is made about physical hardware. (2) **Simplified target-domain actuation.** The Cosserat-rod arm applies idealized per-segment couples and quasi-static settling; it is a harder test domain, not a validated model of a specific physical arm. (3) **Statistical coverage is uneven.** Core results, the shape-conditioning study, the baseline transfer collapse, and the fine-tuning grid carry three-seed statistics; the query-budget, ordering, and guidance experiments use one trained model (with within-experiment pairing across 15–45 targets), the hyperparameter, dataset-size, and morphology studies are likewise single-seed, and the grad-IK reference is a single optimization run per target. (4) **Mode-enumeration hyperparameters** were calibrated for the four-segment arm; morphology results therefore report accuracy and obstacle-task metrics only. (5) **Success-rate granularity** in transfer experiments is 5% ($n_{\text{targets}}=20$).

VIII. CONCLUSION

For the PCC-representable pneumatic continuum arms studied here, multimodal diffusion IK delivers a real, mechanistically attributable advantage—it is redundancy resolution, growing with segment count and vanishing when the target specifies the whole shape—but the learned inverse map itself does not survive a change of dynamics model. Inference-time selection and guidance recover a bounded fraction of

the loss; synthetic gain randomization recovers none; and target-domain fine-tuning buys genuine transfer only at a sharp retention cliff that data mixing cannot smooth. For simulation-trained soft-arm IK, the practical path forward is not better sampling from a source-domain model but mechanisms that let a small amount of target-domain data reshape the model without erasing it.

REPRODUCIBILITY

All code, configuration, per-experiment result files, and the 10^4 -sample Elastica dataset are available at <https://github.com/adhvayjagadeesh/soft-arm-diffusion-ik>.

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